IFAN: An Icosahedral Feature Attention Network for Sound Source Localization

Xin-Cheng Zhu, Hong Zhang, Hui-Tao Feng, Deng-Huang Zhao, Xiao-Jun Zhang, and Zhi Tao

***Abstract*— Currently, sound source localization (SSL) techniques based on deep learning mainly rely on traditional signal processing methods to generate input features. Nevertheless, the applicability of these features in various environments shows significant differences. This study proposes a new single SSL model, called the icosahedral feature attention network (IFAN), to overcome this limitation. The proposed IFAN not only uses steered response power with phase transform (SRP-PHAT) but also develops steered response power with least mean square (SRP-LMS) as inputs of the network. The IFAN network encodes spatial position information into convolution kernels by introducing icosahedral convolutions. In addition, it adaptively learns optimal feature weights based on the input acoustic environment using the sigmoid function to capture the spatial distribution information of the sound source. For single-source SSL and tracking scenarios, the proposed method on the localization and tracking (LOCATA) challenge data corpus outperforms other state-of-the-art models. Moreover, it is capable of learning complementary information even in acoustic simulations involving a wide range of reverberations. The proposed IFAN can thus enhance the robustness and performance in different environments.**

***Index Terms*— Feature attention, icosahedral convolutional neural network (CNN), microphone arrays, sound source localization (SSL), steered response power with phase transform (SRP-PHAT).**

I. Introduction

ADVANCES in artificial intelligence have spawned technologies for human–computer interaction based on intelligent speech. Speech-based interaction is used to transmit semantic information and is an important carrier of information for localizing the sources of sound. Signal processing-based technology for sound source localization (SSL) has been used in a variety of acoustic scenarios, such as for detecting noise in mechanical equipment [1], [2], conference systems [3], [4], intelligent cockpit voice interaction [5], and intelligent robots [6]. SSL methods based on microphone arrays have been extensively researched in the context of signal processing in the past few decades [6]. However, noise, indoor reverberations, and interference by the speaker pose significant challenges to the analysis of acoustic signals. Traditional SSL methods [7], [8], [9], [10], [11], [12], [13], [14] usually first estimate spatial features such as time delay, cross correlation algorithm (GCC), steered response power with phase transform (SRP-PHAT) power maps, interchannel phase/level difference (IPD/ILD), and relative transfer function (RTF). The mapping of features to locations is then established based on the topology of the microphone array under ideal conditions. However, this mapping approach is static such that their performance degrades in environments with variable features as well as those with strong noise and reverberations. Developing a method to precisely localize and track the source of sound in complex scenarios thus remains an outstanding challenge [15].

Deep learning-based SSL methods have been proven to be more robust than traditional methods [16], [17], [18], [19]. A summary of some deep learning-based SSL methods is presented in Table I. Deep learning-based methods use a large amount of training data to learn nonlinear mappings between the signals received by the microphone array and the sound source. Moreover, the relevant data consider prior information regarding noise and reverberations in the environment such that they are applicable to a wide range of scenarios. The input features used for SSL in the network can be the original multichannel signal or its representation related to the localization information, such as the spectrum obtained by the Fourier transform. To retain all the useful information in the original multichannel signals, He et al. [20] used the real and imaginary parts of signals from the time–frequency domain as the input to a network to estimate the direction-of-arrival (DOA) based on convolutional neural network (CNN). Nguyen et al. [21] proposed the spatial cue-augmented log-spectrogram features (SALSA) to detect and localize the sound source. The results of test showed that SALSA outperformed the multichannel log-MEL spectrogram. Cheng et al. [22] proposed a DNN-based model to enhance the difference in the phase of the steering vector and applied it to scenarios involving a variety of noises and conditions of reverberation. Comanducci et al. [23] extracted useful information from multiframe GCC-PHAT features based on CNN and showed that its accuracy was similar to that of SRP-PHAT. An improved version based on GCC, FS-GCC [24], has recently been proposed as well. It can learn information on the time delay under adverse acoustic conditions by using the CNN [25]. Yang et al. [26] directly used the formula for the SRP to construct a spatial spectrum by the phase difference in the direct path as learned by the network. Nguyen et al. [27] calculated the features input to a CNN based on the MUSIC algorithm to estimate the number of

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TABLE I
DEEP LEARNING-BASED SSL

Category	Year	Approach	Input	Output	Network
Signal to location	2018	[31]	magnitude and phase spectrograms	Location identity	CRNN
Feature to Feature	2019	[16]	sine and cosin of IPD	sine and cosin of IPD	Fully Connected
Feature to location	2020	[19]	GCC-PHAT and periodicity degree	Location identity	CNN
Feature to Feature	2020	[23]	GCC	Ray Space Transform	CNN
Feature to location	2020	[25]	Frequency-Sliding GCC	Location identity	CNN
Feature to location	2020	[27]	2D short-time spatial spectrum	Location identity	CNN
Signal to location	2021	[20]	magnitude and phase spectrograms	Location identity	CNN
Feature to Feature	2021	[22]	sine and cosin of IPDs & magnitude spectrograms	sine and cosin of IPD	RNN
Signal to location	2021	[33]	magnitude and phase spectrograms	Location identity	CNN
Feature to location	2021	[34]	SRP-PHAT	Location identity	3D CNN
Feature to location	2022	[21]	SALSA: Log-linear spectrograms & normalized eigenvectors	Location identity	CRNN
Signal to Feature	2022	[26]	magnitude and phase spectrograms	direct-path IPD	CRNN
Feature to location	2022	[28]	circular harmonic features	Location identity	CNN
Feature to Feature	2022	[29]	spherical beamforming map	spherical beamforming map	spherical CNN
Feature to location	2022	[35]	SRP-PHAT	Location identity	spherical CRNN
Feature to location	2023	[36]	SRP-PHAT	Location identity	Icosahedral CNN

sources of sound and the direction of arrival of the short-time spatial pseudo-spectrum. Gong et al. [28] designed circular harmonic features as the input to a frequency-invariant CNN architecture. Most inputs used in research in the area are single features based on the traditional SSL. Examples include inter-channel features, time delay-based features, spectrogram-based features, ambisonic signal representation, and intensity-based features [29]. Such features usually represent the spatial or time–frequency-related information of the signal received by the microphone. However, different features have varying degrees of robustness against indoor environments, because of which deep learning-based networks may not be able to learn all the information of SSL in complex environments from a single feature.

The network architecture used for SSL has generally evolved with the development of deep neural networks in other fields such that the features used are more consistent with the complexity of the scenarios of application. Simpler methods, such as multilayer perceptron, have gradually fallen out of favor, and recent studies have used more complex neural networks. Hirvonen [30] was the first to propose using 2-D CNN for SSL as early as 2015, but this architecture can only classify the orientations of eight spatial regions. Convolutional layers have been proven to be suitable for extracting relevant features of SSL. There are few published works on SSL using only recurrent neural networks (RNNs), as recurrent layers are often combined with convolutional layers to capture the long-term temporal context of spatial information in mobile scenarios. Detection and classification of acoustic scenes and events (DCASEs) challenge adopts a convolutional RNN as the baseline network for the SSL task [31], [32]. However, its accuracy needs to be improved. Krause et al. [33] subsequently investigated different forms of convolutions to this end. The research results found that compared to networks based on 2-D convolutions, complex convolutions, and deep separable convolutions, networks using 3-D convolutions (on the time, frequency, and channel axes) achieved better positioning accuracy but had high computational costs. Diaz-Guerra et al. [34] similarly claimed that 3-D convolutions can more accurately identify the sound source in the presence of reverberations than 2-D and 1-D convolutions. Diaz-Guerra et al. [34] applied 3-D convolutional layers with 1-D causal convolutional layers to equiangular SRP-PHAT power maps to track a single source of sound. The equiangular SRP-PHAT maps computed from compact arrays do not vary with Cartesian translations on isometric projections. Given that the SRP-PHAT power maps of 3-D arrays can be customized to spherical data, Zhong et al. [35] proposed a spherical convolutional RNN (CRNN) based on SRP-PHAT power maps to track a single sound source. This method reduces the processing time and the number of parameters compared with [34], but the accuracy of the SSL system is not further improved. However, the icosahedron is an approximate sphere and is much easier to discretize. Diaz-Guerra et al. [36] claimed that the icosahedral CNN [37] is suitable for learning information on the SSL based on SRP-PHAT power maps and can more accurately identify its position while using fewer parameters than prevalent methods. At present, in the field of deep learning-based SSL, researchers hope to develop models with low complexity, high accuracy, and wide application scenarios.

The authors of this study propose the icosahedral feature attention network (IFAN), in view of the abovementioned challenges to SSL under varying conditions of reverberation. The main contributions of this work can be summarized as follows.

1) We propose steered response power with least-mean-square (SRP-LMS) power maps obtained from reverberation modeling. To explore the complementary contributions of features, both SRP-LMS and classical SRP-PHAT power maps are used as inputs to the network.
2) Inspired by the icosahedral convolution, we propose an IFAN. In contrast to other network structures, it can learn the matching between the input acoustic features and the indoor acoustic environment of the input speech recordings and is provided with lower computational complexity. The proposed network consists of modules for feature extraction, learning the residual features, learning the weights of the features, and fusing them to improve the robustness of single-source scenarios.

II. Icosahedral SRP Power Maps

SRP-PHAT is a delay and beamforming method based on GCC-PHAT calculation. Diaz-Guerra et al. [34], [36] adopted SRP-PHAT maps and GCC-PHAT features for SSL based on deep learning and tested them on open-source databases. The SRP-LMS algorithm proposed here and the traditional SRP-PHAT algorithm have their own advantages in different environments.

A. Classical SRP-PHAT Algorithm

The highly precise calculation of the time delay plays a pivotal role in enhancing the effectiveness of DOA estimation. In an ideal scenario, the model of the signal of the microphone array can be represented as

$$x_m(t) = \alpha_m x(t-\tau_m) + n_m(t) \tag{1}$$

$$x_n(t) = \alpha_n x(t-\tau_n) + n_n(t) \tag{2}$$

where $x(t)$, τ_m, τ_n, α_m, and α_{n2} represent the source signal, the time required for it to reach two microphones, and the attenuation factor of $x(t)$ during transmission to the microphones, respectively; and $n_m(t)$ and $n_n(t)$ denote noise signals.

Controlled beamforming uses the weighting and summation of the sound signals received by a microphone array to form a controllable beam. The beam can be maximally amplified in the direction of the source of sound by adjusting the weights to alter the direction of reception of the array and scanning the entire target space. The spatial filtering used in the case of beamforming can suppress noise and reverberations within the range of the sidelobes. Localization algorithms based on controlled beamforming can be divided into delay-and-sum beamforming and adaptive filtering-and-sum beamforming. Delay-and-sum methods of beamforming use fixed weighting schemes, where the SRP algorithm is commonly used in this context. Methods of adaptive filtering-and-sum beamforming use adaptive weighting schemes and can better adjust to complex environments.

The SRP-PHAT algorithm is a method of SSL that combines PHAT weighting from the TDOA algorithm with the delay-and-sum algorithm from beamforming. Specifically, the SRP-PHAT algorithm divides the spatial region considered to localize the source of sound into multiple subspaces, where the source is assumed to be a point source in each subspace. When the source of sound is located in a particular subspace, the algorithm first estimates the time delay by using the GCC algorithm to obtain the difference in time delays between microphone pairs. It then preprocesses the signal by using PHAT weighting to reduce the influence of noise and reverberations. Following this, it applies beamforming to each subspace and calculates the output power of each beam. The source of sound is localized by comparing the output powers of the beams in all subspaces. This algorithm is robust and has a high accuracy of localization such that it is suitable for use in complex environments.

The controllable response of an array of M microphones for beamforming in the frequency domain is defined as follows:

$$Y(m,\boldsymbol{\theta}) = \sum_{m=1}^{M} G_m(w) X_m(w) e^{jw\tau_m(\boldsymbol{\theta})} \tag{3}$$

where $G_m(w)$ represents the Fourier transform of the filter, $X_m(w)$ represents that of the signal, θ represents the direction vector of the source of sound, and $\tau_m(\boldsymbol{\theta})$ represents the difference in delay between the sound source and the m th microphone relative to a reference microphone.

The power of the controllable response can then be given as

$$P(\boldsymbol{\theta}) = \int_{-\infty}^{+\infty} |Y(m,\boldsymbol{\theta})|^2 d = \int_{-\infty}^{+\infty} Y(m,\boldsymbol{\theta}) Y(m,\boldsymbol{\theta})^* dw \tag{4}$$

where $*$ denotes the complex conjugate. Also, $R_{x_m x_n}$ is the GCC function for the mth and nth microphone signals

$$R_{x_m x_n}\left(\tau_{m,n}(\boldsymbol{\theta})\right) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} Y(m,\boldsymbol{\theta}) Y(m,\boldsymbol{\theta})^* dw \tag{5}$$

where $\tau_{m,n}(\boldsymbol{\theta}) = \tau_m(\boldsymbol{\theta}) - \tau_n(\boldsymbol{\theta})$.

Although the direct implementation of (4) is computationally intensive, it can be computed using (5). The output power $P(\boldsymbol{\theta})$ can then be expressed as

$$P(\boldsymbol{\theta}) = 2\pi \sum_{m=1}^{M} \sum_{n=1}^{M} R_{x_m x_n}\left(\tau_{m,n}(\boldsymbol{\theta})\right). \tag{6}$$

Furthermore, we ignore the constant coefficients 2π that do not influence direction estimation. Also, formula (6) combined with the PHAT transformation $G_m(w) = 1/|X_m(w)$, the SRP-PHAT algorithm can be obtained as follows:

$$P^{\text{PHAT}}(\boldsymbol{\theta}) = \sum_{m=1}^{M} \sum_{n=1}^{M} R_{x_m x_n}^{\text{PHAT}}\left(\tau_{m,n}(\boldsymbol{\theta})\right). \tag{7}$$

B. SRP-LMS Algorithm

The least-mean-square (LMS) time delay estimate algorithm was proposed by Feintuch et al. [38] in 1981 and is based on the concept of adjusting the coefficients of the filter according to the similarity between the received signals. The minimum mean-squared error is used to cause one received signal to converge toward another. Once the peak value of the optimal filter has been estimated, the time delay between the signals can be extracted from the coefficients of the filter. This algorithm uses a model of reverberation and is commonly used for echo cancellation. It has a superior ability to suppress reverberations than the GCC method. Nevertheless, the indoor environment is often significantly more complex than ideal reverberation conditions, thereby rendering the LMS-based estimation of the time delay susceptible to errors.

The received signals in the model of the array of microphones are denoted by $x_1(n)$ and $x_2(n)$, with $x_1(n)$ serving as the input signal. The objective of the algorithm is to ensure that the output signal $x_2(n)$, obtained after passing $x_1(n)$ through the filter $h(n)$, infinitely approaches the target signal $x_2(n)$

$$\hat{x}_2(n) = x_1(n) * h_{\text{opt}}(n) \tag{8}$$

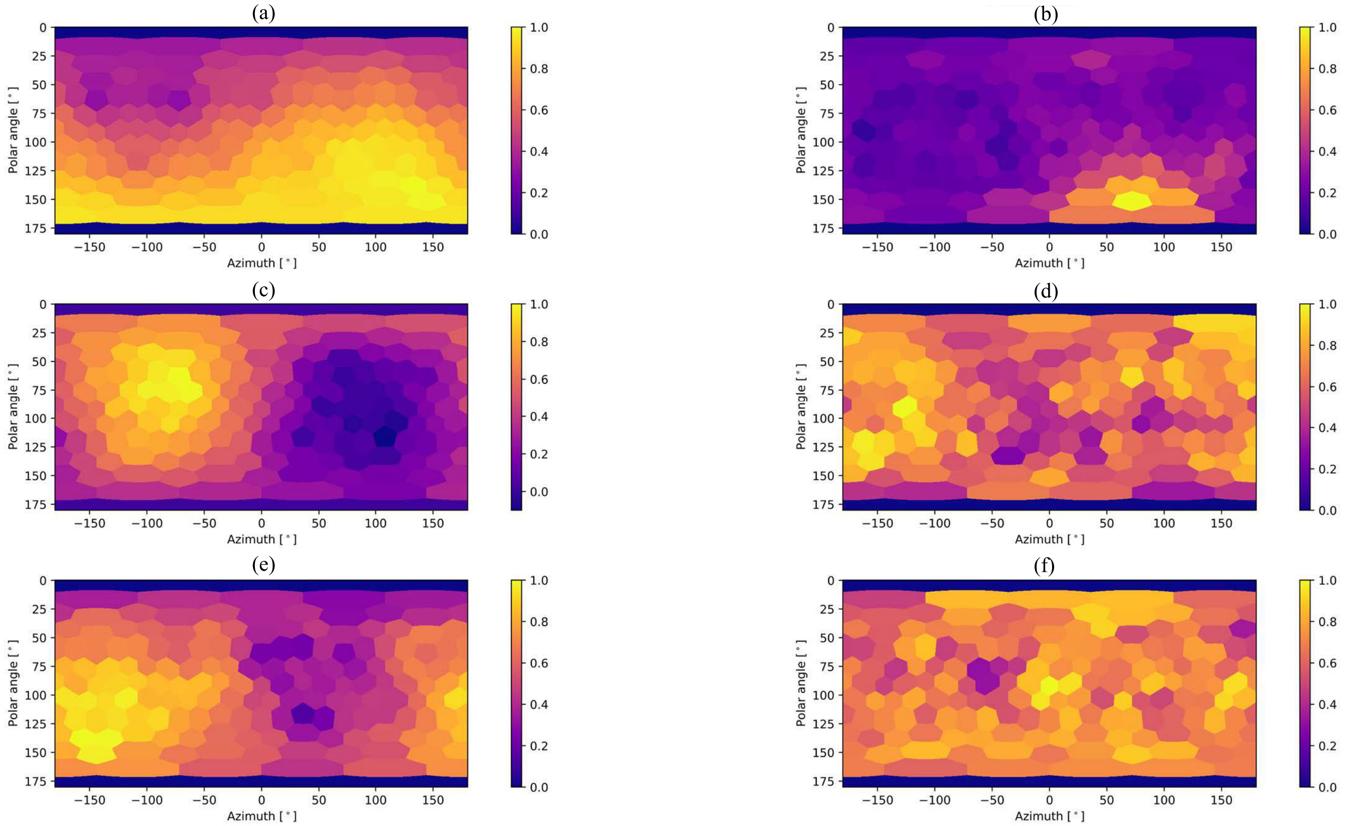

Fig. 1. Examples of icosahedral SRP-PHAT power maps and icosahedral SRP-LMS power maps with resolution $r = 2$ in different indoor acoustic scenarios. (a) Icosahedral SRP-LMS power maps (SNR = 30 dB, T60 = 0.2 s), (b) icosahedral SRP-PHAT power maps (SNR = 30 dB, T60 = 0.2 s), (c) icosahedral SRP-LMS power maps (SNR = 30 dB, T60 = 0.8 s), (d) icosahedral SRP-PHAT power maps (SNR = 30 dB, T60 = 0.8 s), (e) icosahedral SRP-LMS power maps (SNR = 5 dB, T60 = 0.8 s), and (f) icosahedral SRP-PHAT power maps (SNR = 5 dB, T60 = 0.8 s).

where $h_{\text{opt}}(n)$ is the optimal filter

$$h_{\text{opt}}(n) = E\{x_2(n), x_1(n) * h(n)\}. \tag{9}$$

When the minimum mean-squared error is considered, the peak value of the filtering coefficient $H(n)$ corresponds to the time delay

$$\tau = \arg\max_n h_{\text{opt}}(n). \tag{10}$$

The LMS algorithm uses a model of reverberations that contains useful information on the DOA under a low signal-to-noise ratio (SNR) and large reverberations. We use GCC-PHAT to calculate the SRP-PHAT as a reference and apply the LMS algorithm to calculate the optimal coefficients of the filter between each pair of microphones in the array. These coefficients are called SRP-LMS after summation. The SRP-LMS features can be obtained by summing the coefficients containing information on the time delay

$$P^{\text{LMS}}(\boldsymbol{\theta}) = \sum_{m=1}^{M}\sum_{n=1}^{M} h_{\text{opt}}\big(\tau_{m,n}(\boldsymbol{\theta})\big). \tag{11}$$

However, reverberations in indoor settings are often much more complex than assumed under ideal conditions. Therefore, the LMS-based estimation of the delay is prone to errors. We propose using dual-channel features to solve this problem. SRP-PHAT and SRP-LMS maps are used as the input to the network. In the case of spherical coordinates, formulas (7) and (9) are used to map the icosahedral grid and obtain the corresponding icosahedral SRP power maps.

Fig. 1 shows that the canvas parameter, cmap, uses plasma. The canvas appears purple at a low power output, and the maximum output power should appear in yellow. The subplots for each row were drawn from signals received from multiple channels and generated by simulations in a room of the same size as well as the same SNR, microphone, and trajectory of the source of sound. The SNRs of the subplots in Fig. 1(a) and (b) were 30 dB, and T60 was 0.2 s. Icosahedral SRP-PHAT maps can accurately calculate the direction of the source of sound in an environment with low reverberations and a high SNR. Although the features in both exhibited peaks of the output power at a high SNR, the single peak of the icosahedral SRP-PHAT was more prominent than that of the icosahedral SRP-LMS. The latter performed poorly due to miscalculations. The SNR of the subplots shown in Fig. 1(c) and (d) was 30 dB, and T60 was 0.8 s. The icosahedral SRP-PHAT incorrectly calculated a large number of peaks of output power when the coefficient of reverberation T60 increased sharply to 0.8 s, while the icosahedral SRP-LMS demonstrated its advantages under large reverberations. However, Fig. 1(e) and (f) shows that that SRP-LMS was also affected when the SNR was 5 dB and T60 was 0.8 s.

III. Icosahedral CNNs

Traditional methods based on the CNN cannot handle tasks involving 2-D planes, such as those on 2-D manifolds. A typical 2-D manifold is a sphere. Traditional CNNs encounter distortion or suffer from a loss of information when processing

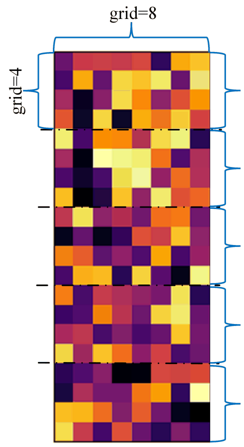

Fig. 2. Icosahedral grid ($r = 2$).

spherical data, whereas icosahedral CNNs can better preserve such data. Distortion changes the representation of images and deforms the local content, where this violates translational invariance: a key property of the signals required for convolutional operations. Therefore, any attempt to represent spherical images as planar images leads to some degree of distortion. Cohen proposed a method for performing a gauge-equivariant convolution on an icosahedron. Icosahedral CNNs offer a highly practical alternative to spherical CNNs because they are capable of implementing gauge-equivariant convolutions with a single conv2d function call.

The icosahedron is among the most uniform and accurate discretizations of a sphere. A spherical mesh can be obtained by gradually subdividing each face of a unit icosahedron into four triangles and reprojecting each node to a unit distance from the origin. This method of discretization preserves the geometric shape of the sphere such that the geodesic distance between any pair of discretized nodes is nearly constant. This simplifies the learning of lifting operators and enables weight sharing. Representing spherical images as subdivided icosahedra reduces spherical distortion and improves the accuracy of CNNs compared with techniques that operate on equirectangular images.

The icosahedral CNN is a highly symmetric CNN that can handle data with spherical symmetry and process input data in multiple directions, thereby enhancing the capability of representation of the data. In the context of SSL, sound waves propagate in the air in a spherically symmetric manner, because of which the signals of sound from it can be represented as functions on a sphere. Icosahedral CNNs can process these functions on a sphere to make them suitable for SSL. Moreover, icosahedral CNNs can use the fast Fourier transform (FFT) when processing data on a sphere to reduce computational complexity and the consumption of memory, thereby improving the efficiency of processing. Icosahedral CNNs can better handle complex acoustic environments than traditional methods of SSL and can adaptively adjust their precision.

The icosahedral grid is constructed from a series of refined triangular grids. It consists of 20 equilateral triangular faces,

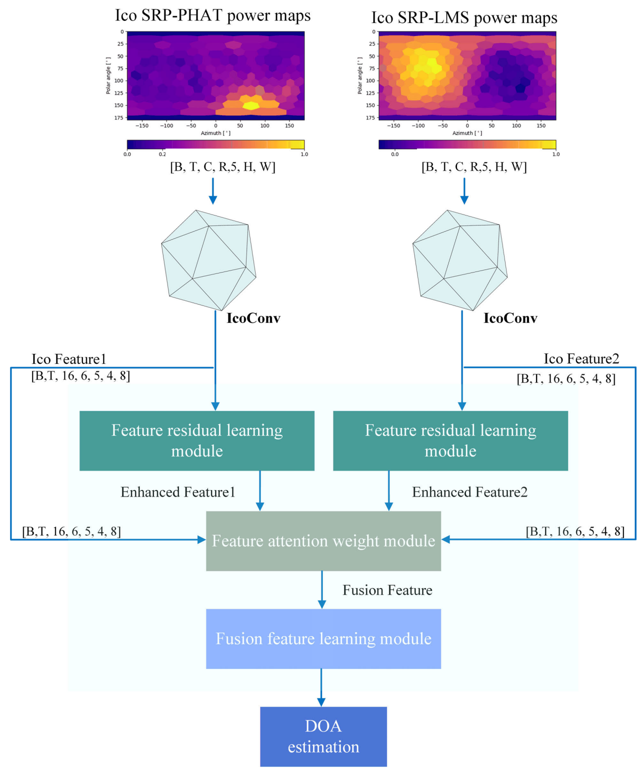

Fig. 3. System framework of IFAN.

30 edges, and 12 vertices. Starting from the vertices, three new points are introduced in each equilateral triangular face to evenly subdivide the grid into four smaller equilateral triangular faces. By repeating this process r times, we obtain a grid with $5 \times 2^{2r+1} + 2$ points. Five charts are generated in this process. Finally, the icosahedral grid maps to a rectangular grid of $5 \times \cdot 2^r \times 2^{r+1}$, as shown in Fig. 2. The vertex of the icosahedral grid has five neighbors, and the remaining internal sampling points have six neighbors each. They are called pentagonal and hexagonal pixels. Thus, the hexagonal kernel of the 3×3 2-D kernel is used to implement an icosahedral convolution. Unlike the traditional convolution, the hexagonal nuclei of icosahedral convolutions need to use six channels to consider their six possible rotations, where this avoids a limit on the isotropy of the core. An icosahedral convolution is achieved by a single conv2d function calling G-Padding and the kernel extension. The specific algorithm is as follows:

$$GConv(f, w) = \text{conv2d}(GPad(f), \text{expand}(w)). \quad (12)$$

IV. Proposed Method

A. Model Architecture

We propose the IFAN for SSL. Its framework is shown in Fig. 3. The input to the IFAN model consists of icosahedral SRP-PHAT maps and SRP-LMS maps. These are seven dimension tensors, with dimensions $B \times T \times C \times R \times 5 \times H \times W$, where B is the batch size, T is time, C represents the channels, and R denotes the six channels required for the hexagonal nucleus of the icosahedral convolution. Its initial value for scalars of the input feature is one. H and W are the

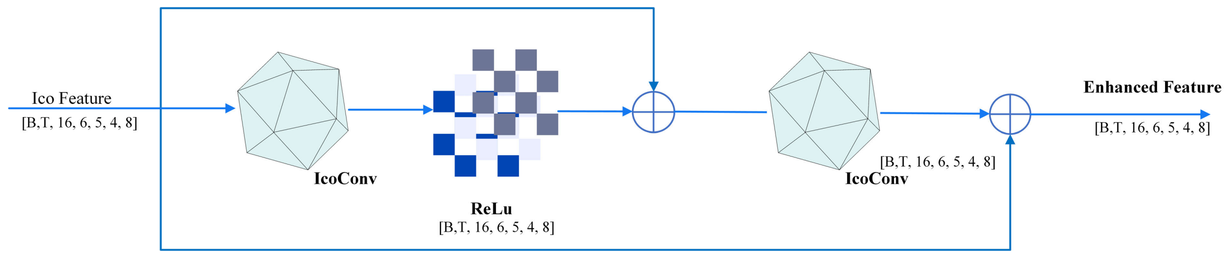

Fig. 4. Feature residual learning module.

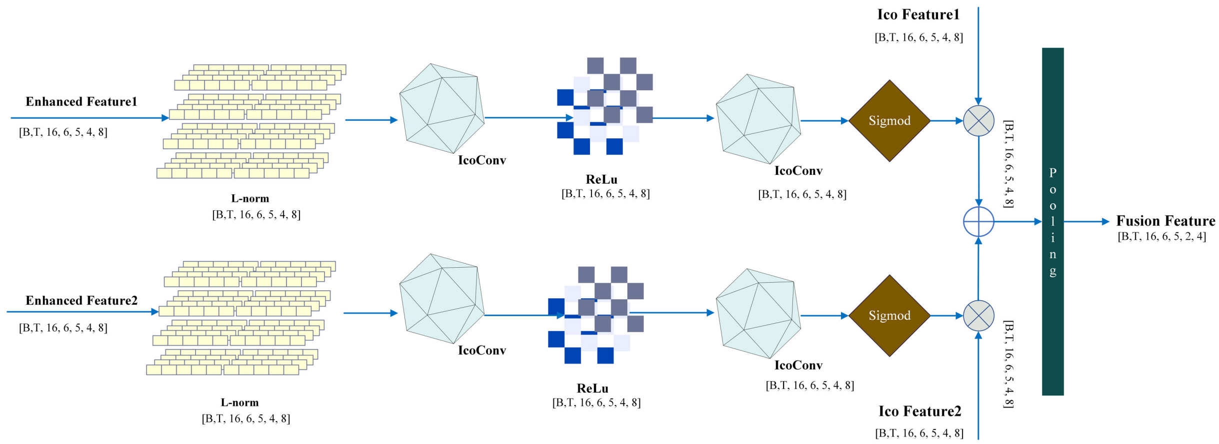

Fig. 5. Feature attention weight module.

length and width of the icosahedral grid of the input feature, respectively. The input icosahedral features convolved by an icosahedral layer are called Ico SRP-PHAT power maps and Ico SRP-LMS power maps, respectively.

The IFAN also contains modules to learn the residual features, their attention weights, and fusion. The dimensions of their channels are extended to 16, and the regular feature R has a value of six. This step is used for feature extraction. The structure of the module to learn the residual features is shown in Fig. 4.

The module to learn the residual features is mainly composed of convolutional layers, ReLU activation layers, and a residual structure. To adjust dimensional information contained in the tensor of input features (Ico Feature1 and Ico Feature2), we use the icosahedral convolution as the convolution layer in this module. When the traditional convolutional structure deepens the network, valuable information is lost. The module to learn the residual features not only deepens the network and reduces the loss of information pertaining to the input features but also expedites convergence. Specifically, it can improve the ability of the features to characterize the DOA, enhance information on the input features of the source of sound in the icosahedral grid, retain potentially valuable information, and suppress unimportant information.

As shown in Fig. 5, this module consists of an icosahedral normalization layer (L-norm), a convolutional layer, a ReLU activation function (ReLU), a sigmoid activation function (sigmoid), and a pooling layer (pooling). The L-norm helps accelerate convergence and improve the robustness of the model. The SRP-LMS and SRP-PHAT have varying advantages and disadvantages in different indoor environments. The enhanced features are used to obtain the coefficient of weight of each input feature under various environmental conditions by training the module used to assign attention weights to them. To obtain this coefficient, the icosahedral normalization layer is first applied to the enhanced features to obtain the corresponding normalized features $L-f$ on the dimensions of the input channel (C) and the icosahedral convolutional channel (R)

$$L-f = \text{LNormIco(enhanced feature)} \tag{13}$$

where LNormIco is the icosahedral normalization layer.

Convolutional layers are used to further enhance the ability of the features to represent the location of the source of sound. The adaptive coefficient of the weights w of the two types of features in different environments is combined with the sigmoid layer and then fed back

$$w = \text{Sigmoid}(\text{Ico}conv(\text{ReLU}(\text{Ico}conv(L-f)))) \tag{14}$$

where w is the coefficient of weight, sigmoid is the sigmoid activation function, ReLU is the ReLU activation function, and Ico$conv$ is the icosahedral convolution.

w is multiplied with the corresponding output of the module to learn the residual features to obtain adaptive features. The latter, as obtained from the dual-channel features, are then fused. The resulting feature has the same number of dimensions as icosahedral maps with $r = 1$. In this way, the receptive field of the output convolutional neuron of the IFAN covers all information on the input icosahedral SRP-LMS maps and icosahedral SRP-PHAT maps, with resolutions of

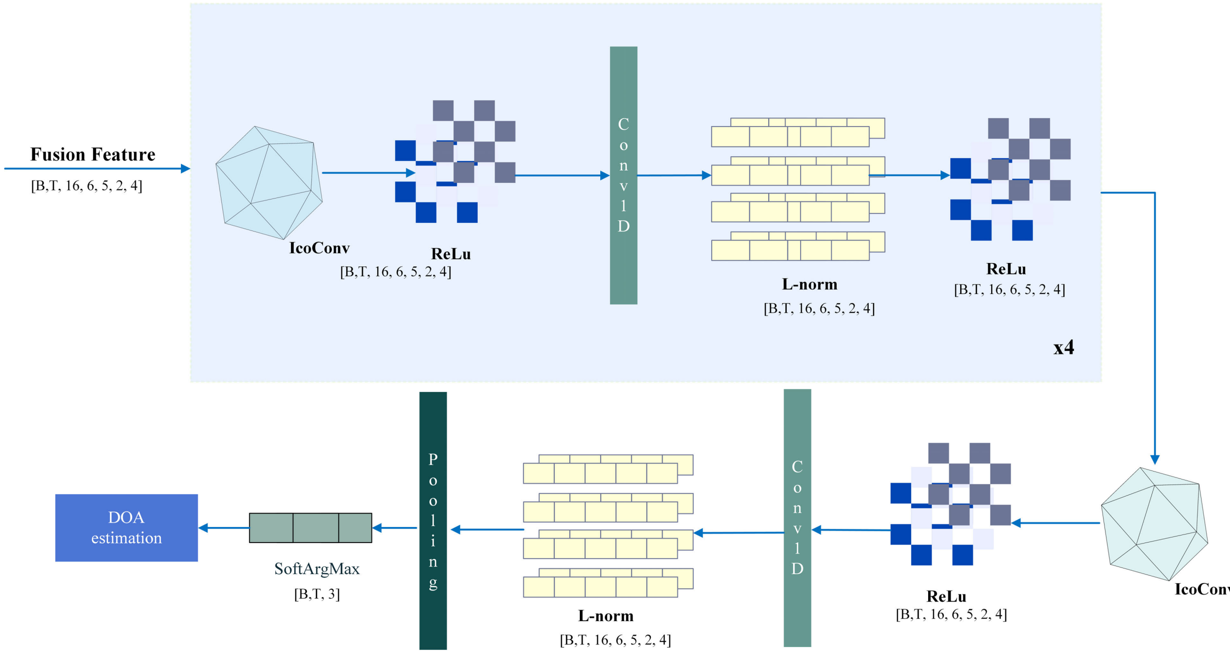

Fig. 6. Fusion feature learning module.

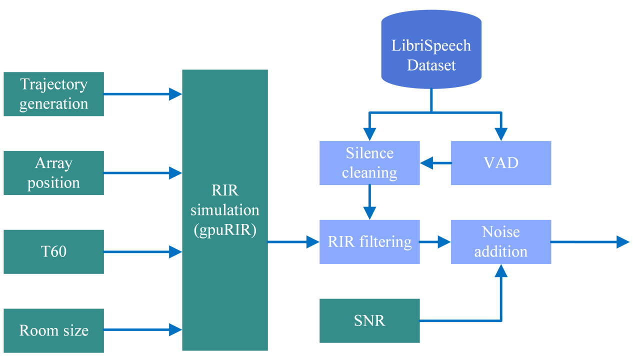

Fig. 7. Dataset generation process.

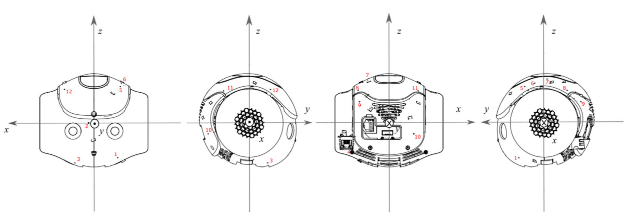

Fig. 8. Geometries of the 3-D robot-head microphone array.

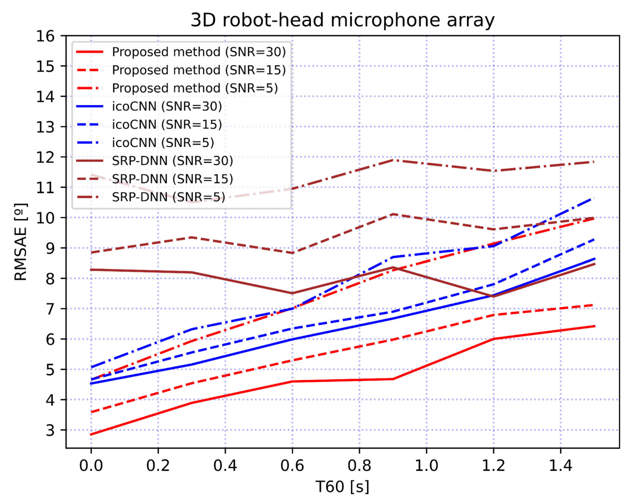

Fig. 9. Localization root-mean-squared angular error for several SNRs and reverberation times.

$r = 1$ and 2. As shown in Fig. 6, in the fusion feature learning module, the dimension of the fusion feature is $B \times T \times C \times R \times 5 \times 2 \times 4$. The icosahedral convolution and 1-D convolution further enhance the robustness of the model to the position of the sound source. The feature fusion module combines the icosahedral convolution of 32 nuclei and the 1-D time convolution to learn the information contained in the previous frame and finally obtains static and dynamic DOA information.

B. Model Training

To verify the comparative effectiveness of the proposed method, we trained its model consistently with the procedure used in [34] and [36]. Fig. 7 shows a framework for generating simulation data to facilitate model training. LibriSpeech [39] utterances serve as the sound sources for simulation data. In the LibriSpeech corpus, the LibriSpeech train-clean-100 subset containing 100 h of English speech data is used for simulating data. The background noise of the sound source used to simulate the signal may strengthen the model to learn the location information of the background noise. To avoid the influence of possible background noise in the Librispeech utterances on localization and tracking (LOCATA) in silent segments, we use a speech activity detector to detect silent segments and completely remove the signal in those frames. We generated random source trajectories and applied RIR simulations [40] by using an image source method [41] at a sampling rate of 16 kHz, which required setting the SNR and the duration of the reverberations. IEEE-AASP Sound Source LOCATA Challenge employed a 3-D robotic head microphone

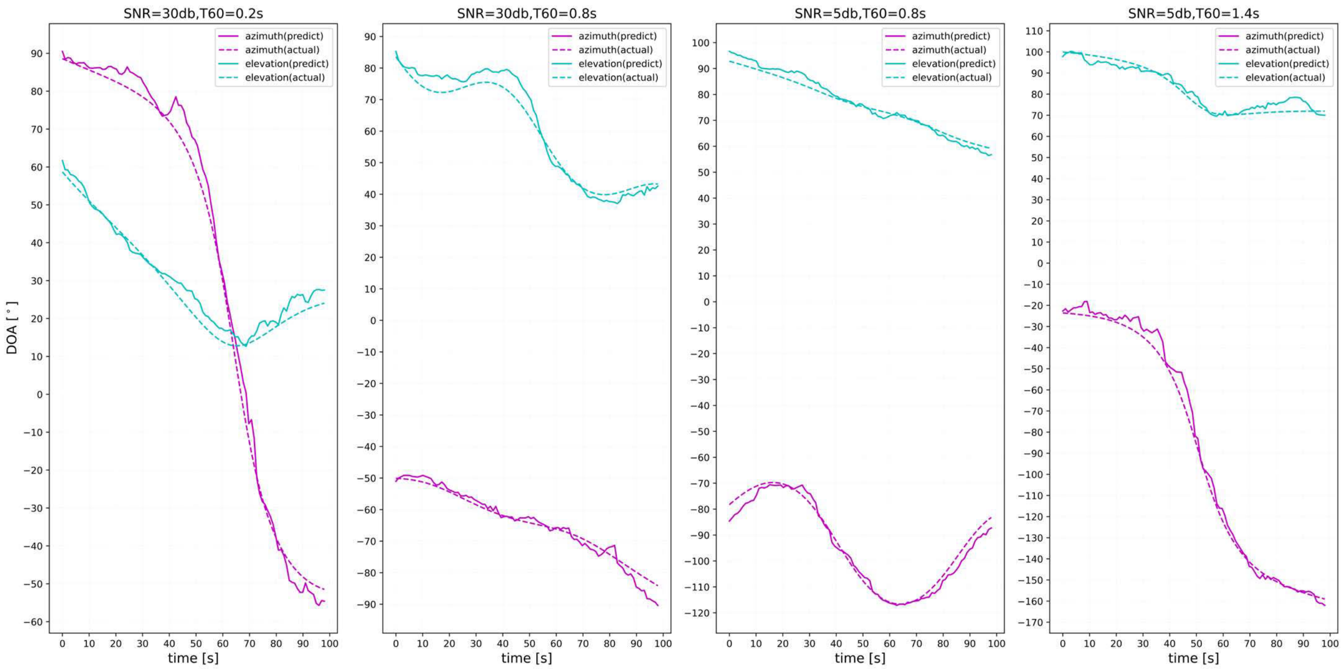

Fig. 10. Examples of the estimated DOA by the proposed method in the simulations.

array [42], [43], with a minimum and maximum microphone-to-microphone distance of 1.3 and 12.1 cm, respectively. It is worth noting that the simulation data used for training the IFAN network adopt the same topology structure as the 3-D array of LOCATA, as illustrated in Fig. 8. We focused on single-source localization tasks and set the track of source at a duration of 20 s as a random state of motion. Microphone arrays were randomly placed in rooms with dimensions ranging from $3 \times 3 \times 2.5$ to $10 \times 8 \times 6$ m^3. However, the microphone array was localized to 10% of the size of the room in both the horizontal and the vertical directions, and its height was limited to the lower half of the room. Omnidirectional Gaussian noise was used, with an SNR distributed randomly and evenly between 5 and 30 dB. For the reverberation time, we selected T_{60} values that varies randomly from 0.2 to 1.3 s during the whole training. We set the frame length to 4096 and the frameshift to 3072 and then calculated the icosahedral SRP-PHAT maps and icosahedral SRP-LMS maps for each. The mean squared error in the spatial Cartesian coordinates (x, y, z) was used as the loss function for the 3-D array because this metric attends to the angular distance between spherical coordinates. The Adam algorithm was to train the model over 80 epochs. Each epoch contains 585 trajectories (the number of book chapters in the LibriSpeech train-clean-100 subset) with trajectories of 20 s. The coarse learning strategy was used, and the SNR for the first 20 epochs was set to 30 dB. The batch size was one and the learning rate was 1e$-$4. The SNR for epochs after the first 20 was chosen randomly in the range of 5–30 dB. The batch size was then increased to 10 and the learning rate was reduced to 1e$-$5.

V. Experimental Results

A. Evaluation Metrics

The great-circle distance ($\Delta\delta_i$) between the predicted DOA angle and the actual DOA angle in the spherical coordinate system can be expressed as follows:

TABLE II

LOCATA Challenge Tasks and the Test Scheme Used in This Study

	Sound source	Microphones array	Talker	Whether to challenge
Task1	single	static	static	Yes
Task2	multiple	static	static	No
Task3	single	static	moving	Yes
Task4	multiple	static	moving	No
Task5	single	moving	moving	Yes
Task6	multiple	moving	moving	No

$$\Delta\delta_i = \cos^{-1}\left(\cos\emptyset_p \cos\emptyset_a + \cos\left(\vartheta_p - \vartheta_a\right)\sin\emptyset_p \sin\emptyset_a\right) \quad (15)$$

where $\Delta\delta_i$ represents the spherical distance difference in the ith frame, $\emptyset_p$ is the predicted elevation angle of the DOA, and $\emptyset_a$ is the actual elevation angle of the DOA. At the same time, ϑ_p is the predicted azimuth angle of the DOA and ϑ_p is the actual azimuth angle of the DOA.

We used the root-mean-squared angular error (RMSAE) between the estimated and the actual DOA as the evaluation metric in this study. The RMSAE is calculated as

$$\text{RMSAE} = \frac{180}{\pi}\sqrt{\frac{\sum_{i=1}^{n}(\Delta\delta_i)^2}{n}} \quad (16)$$

where n is the number of samples.

B. Experimental Evaluation

1) Evaluation Using Simulated Data: To delve into the performance of the proposed model under various acoustic conditions, we first evaluated our model using synthesized signals simulated using the same steps as the training dataset. Also, the LibriSpeech test-clean subset containing 100 h of

TABLE III

RMSAE [∘] OF THE DOA ESTIMATED FOR THE LOCATA DATASET WITH THE IFAN AND BASELINE METHODS (THE NUMBER OF FRAMES CONTAINING THE SOUND SOURCE WAS SILENT)

Model	FLOPs [MAC]	Params. [M]	Input features		Task1 (13)	Task3 (5)	Task5 (5)	Standard deviation (23)	Median (23)	Average (23)
SELD net [31]	0.55G	5.34	magnitude and phase spectrograms		20.84	26.51	25.25	12.33	20.56	23.03
Cross3D [34]	7.77G	21.35	Equiangular SRP-PHAT maps		5.28	9.92	12.49	5.71	5.97	7.86
SRP-DNN [26]	3.81G	0.86	log-magnitude and phase spectrograms		6.46	9.66	7.40	4.50	6.20	7.36
Icosahedral CNN [36]	24.95M	0.19	Icosahedral SRP-PHAT maps	r=1	8.04	10.53	15.48	5.00	9.91	10.20
	74.47M	0.29		r=2	5.88	8.94	11.13	3.53	7.17	7.69
	0.27G	0.39		r=3	5.57	10.07	11.54	3.82	6.66	7.85
	1.06G	0.48		r=4	5.25	9.57	10.95	3.51	6.60	7.43
IFAN	6.36M	0.07	Icosahedral SRP-PHAT maps& Icosahedral SRP-LMS maps	r=2	3.88	8.50	10.01	4.19	5.44	6.22

TABLE IV

RMSAE [∘] OF THE DOA ESTIMATED FOR THE LOCATA DATASET WITH THE IFAN AND BASELINE METHODS (THE NUMBER OF FRAMES CONTAINING THE SOUND SOURCE WAS NOT SILENT)

Model	FLOPs [MAC]	Params. [M]	Input features		Task1 (13)	Task3 (5)	Task5 (5)	Standard deviation (23)	Median (23)	Average (23)
SELD net [31]	0.55G	5.34	magnitude and phase spectrograms		18.35	24.13	22.39	10.32	17.88	20.49
Cross3D [34]	7.77G	21.35	Equiangular SRP-PHAT maps		5.48	8.86	10.73	5.37	5.74	7.36
SRP-DNN [26]	3.81G	0.86	log-magnitude and phase spectrograms		6.99	4.48	5.16	3.08	4.91	6.05
Icosahedral CNN [36]	24.95M	0.19	Icosahedral SRP-PHAT maps	r=1	8.34	8.97	13.03	4.17	9.94	9.50
	74.47M	0.29		r=2	6.08	7.29	8.87	2.40	6.79	6.97
	0.27G	0.39		r=3	5.78	6.93	9.66	2.69	6.53	6.87
	1.06G	0.48		r=4	5.23	7.87	8.49	2.26	6.45	6.51
IFAN	6.36M	0.07	Icosahedral SRP-PHAT maps& Icosahedral SRP-LMS maps	r=2	3.76	5.90	6.59	2.67	4.66	4.84

English speech data is used as the sound source for simulation data. The assessment method in this article did not take into account the data of the first five frames due to the artificial influence on the results from the silence of some utterances contained within a short period of time.

Diaz-Guerra et al. [36] claimed that a resolution for icosahedral mapping higher than $r = 2$ cannot improve the accuracy of DOA estimation efficiently. Input feature maps at $r = 2$ may contain all the useful information for localization. We thus used this setting, as shown in Fig. 9. To analyze the robustness of the proposed method, we generated the corresponding simulation scenes for several specific values of T_{60} and SNR for testing. It can be clearly seen that the proposed IFAN model has a better performance than Icosahedral CNN [36] and SRP-DNN [26] models, especially when the reverberation coefficient is at a moderate level. Even in extreme cases of low SNR and strong reverberation, such as a SNR ratio of 5 dB and a reverberation coefficient T_{60} of 1.4 s, our proposed model IFAN still exhibits good performance.

To illustrate this point more clearly, Fig. 10 shows the examples of simulation results for the proposed method in four scenarios. It was tested on speech samples with low reverberation level and low noise (SNR = 30 dB and T_{60} = 0.2 s), higher reverberation level and low noise (SNR = 30 dB and T_{60} = 0.8 s), higher reverberation level and high noise (SNR = 5 dB and T_{60} = 0.8 s), and high reverberation level and high noise (SNR = 5 dB and T_{60} = 1.4 s). The solid line represents the predicted DOA angle, while the dashed line represents the actual DOA angle. The results of sound source tracking at both azimuth angle and elevation angles of the DOA are displayed on each subgraph. Even when the reverberation coefficient T_{60} is 1.4 s, the system can effectively track moving sound sources, and the error of DOA azimuth and elevation is within $\pm 10^\circ$.

2) Evaluation Using the LOCATA Dataset: The accuracy of localization of the source of sound is affected by many factors in actual acoustic environments, including indoor reverberation, noise, interference, and periods of silence.

The sound source thus needed to be continuously recorded to assess the performance of the proposed algorithm. We used the LOCATA dataset, which contains empirical data, to this end. The database was created based on recordings at the Computing Laboratory, Humboldt University, Berlin. The size of the laboratory was $7.1 \times 9.8 \times 3$ m^3 and the duration of reverberations was 0.55 s. The database used OptiTrack, an optical tracking system, to synchronize the live localization data.

Data obtained from a pseudo-spherical array of 12 microphones, shown in Fig. 8, were used to test the proposed method as well. The challenge consisted of six static and dynamic tasks, as shown in Table II. The proposed algorithm was applicable only to tasks involving a single source, because of which it was tested only on Tasks 1, 3, and 5. The scenario in Task 1 involves a single static sound source, which helps to investigate in detail the adverse effects of reverberation and noise on SSL. Using the data from Task 3, it is possible to study the source directionality and the impact of head and body rotation on human speakers. The fully dynamic scenario of Task 5 aims to explore more complex and practical scenarios. We made sure that the test dataset was consistent with the dataset used for Cross3D [34] and Icosahedral CNN [36]. “Task1” in LOCATA’s “eval subset” contained 13 voice recordings, while “Task3” and “Task5” each contained five recordings. Table III compares the average

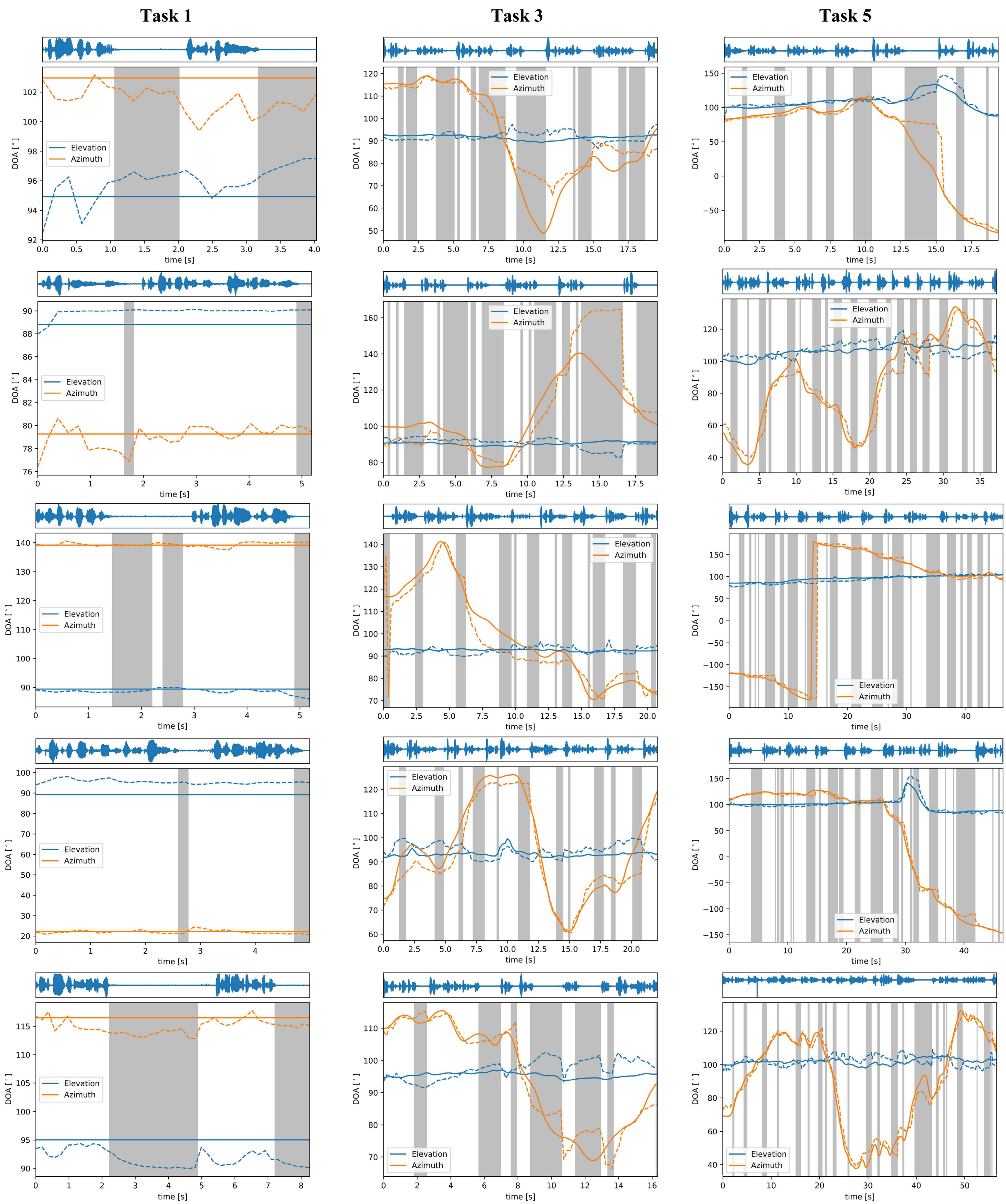

Fig. 11. Examples of DOA estimation for the first five recordings of each single-source task in the LOCATA dataset.

RMSAE [∘] of the DOA obtained by the icosahedral CNN [33] and cross3D models [34]. The input to the cross3D model consisted of 2-D equiangular SRP-PHAT maps. Owing to the low resolution of these 2-D equiangular maps, we compared only the highest values of precision obtained by this method. Diaz-Guerra et al. [36] reported the results of icosahedral SRP-PHAT maps with different resolutions and concluded that a resolution of mapping higher than $r = 2$ does not improve the accuracy of DOA estimation. We thus set $r = 2$ here. Table IV shows the RMSAEs in the DOA estimated by all the methods without considering the frames in which the source of sound was silent.

We will compare the proposed model with four deep learning-based SSL models in terms of localization accuracy, computational complexity, and model complexity. Calculate their RMSAE, floating point operations (FLOPs), and parameters (Params.) indicators separately. The comparison of the results in Tables III and IV shows that the proposed model has the best average accuracy and lowest computational complexity and is superior to other deep learning-based models. In the case where the sound source contains silent frames, the proposed method achieves an average improvement of 69.64%, 20.87%, 15.49%, and 16.29% in accuracy compared to these four models in these three tasks. In the case where the sound source does not contain silent frames, the proposed method achieves an average improvement of 76.38%, 34.24%, 20%, and 25.65% in accuracy compared to these four models. IFAN uses icosahedral convolution to encode spatial position information into the convolution kernel. The model introduces SRP-LMS and SRP-PHAT as inputs. The feature attention weight framework proposed by IFAN can adaptively learn feature weights, better capture the spatial distribution information of sound sources, and improve the positioning accuracy and robustness of the model. However, the performance of the model still highly depends on the quality and diversity of input features. If the quality of input features is poor, it may affect the performance of the model. However, it is also more robust than simply inputting features. The CRNN network used by SELD net [31] performs poorly in mapping spectral maps to sound source locations. The network structure of Cross3D [34] is relatively complex with many parameters, requiring more time and computational resources for training and inference. IFAN is fully ahead of all three models except for SRP-DNN [26] in the LOCATA database. However, the effectiveness of SRP-DNN in simulating data is not as outstanding. In the scenario of moving sound sources, SRP-DNN seems to exhibit better performance. This may be due to the RNN effect used by SRP-DNN, but it also reduces its static SSL effect. It is worth pointing out that SRP-DNN can detect the location of multiple sound sources through iterative processing after estimating the direct path IPD. In short, its robustness and computational complexity are not as good as our proposed method.

Fig. 11 shows the results of the DOA estimated for the first five recordings of each single-source task in the LOCATA dataset by the models. The solid lines represent the ground-truth DOA and the dashed lines represent that estimated by the proposed method based on input maps at a resolution of $r = 2$. In a silent frame, the result of model estimation may offset the correct DOA value, which affects the estimation accuracy of the next nonsilent frame. In particular, this error may be larger when the sound source is moving or the microphone array is moving. However, the estimation of the proposed model always remains close to the actual DOA of the source even in silent frames. This further indicates that a combination of SRP-PHAT and SRP-LMS maps as input features can be used to train a robust model of SSL.

VI. Conclusion

In this study, the authors proposed the IFAN based on icosahedral CNNs to robustly localize and track the source of sound. The network takes as input icosahedral SRP-PHAT and SRP-LMS power maps that are obtained through reverberation modeling.

The results of simulations by using two different structures of the microphone array showed that the proposed method outperformed state-of-the-art techniques in terms of accuracy, especially in the case of large reverberations. The icosahedral SRP-PHAT and SRP-LMS power maps are suitable as input to systems of SSL. The proposed model can be applied to scenarios involving mobile sound sources and can effectively perform even on challenging datasets containing actual recorded data.

However, because the proposed IFAN cannot solve the problem of localizing multiple sources of sound, it is not suitable for complex indoor environments. We plan to explore other techniques, including frame-wise permutation-invariant training, to improve its robustness in future work in the area.

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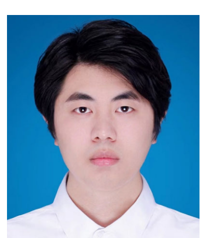

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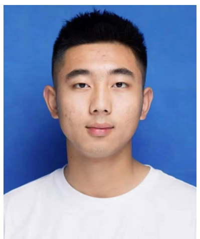

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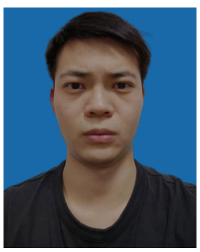
His current research interests include speech signal processing.



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